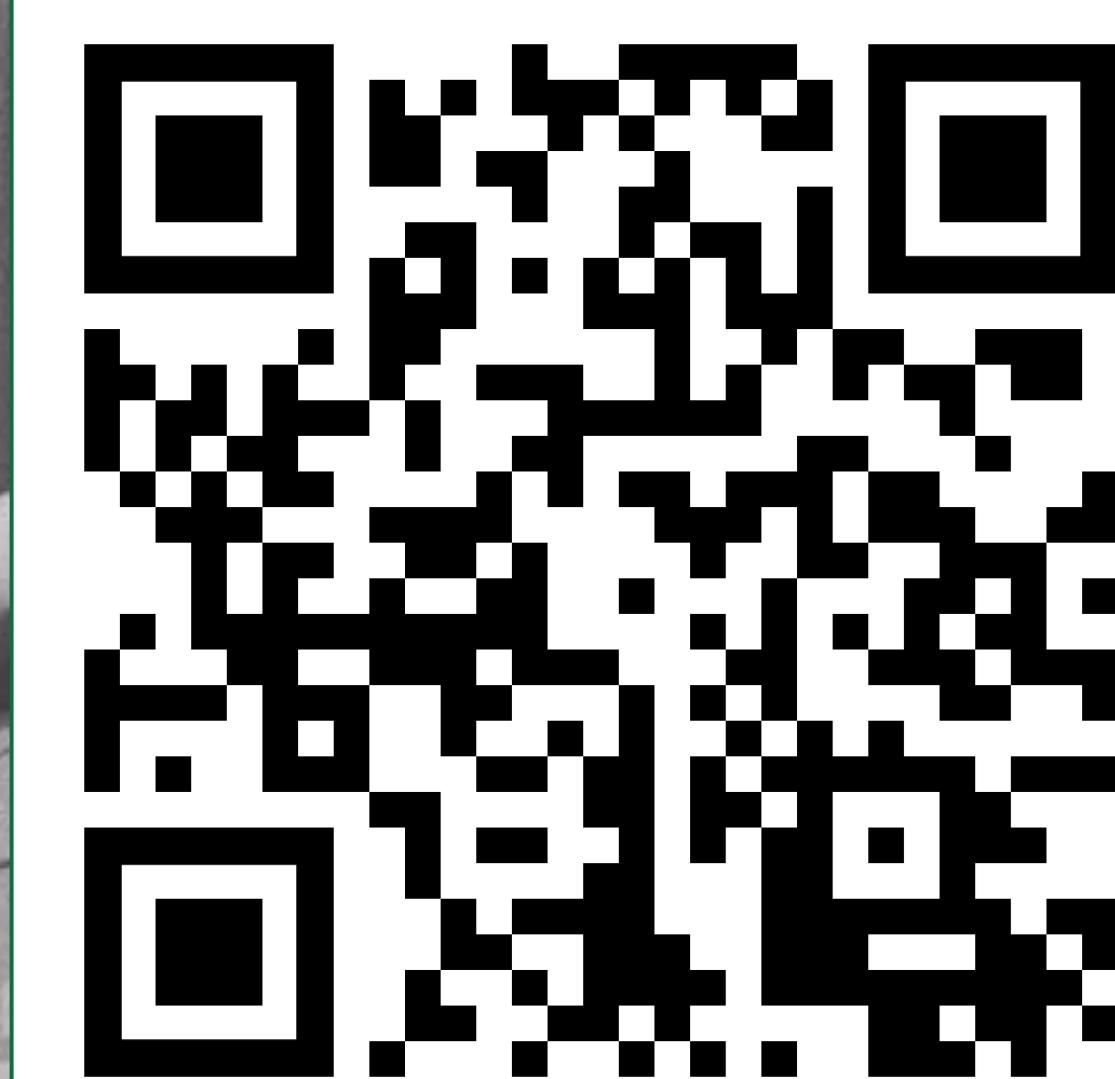
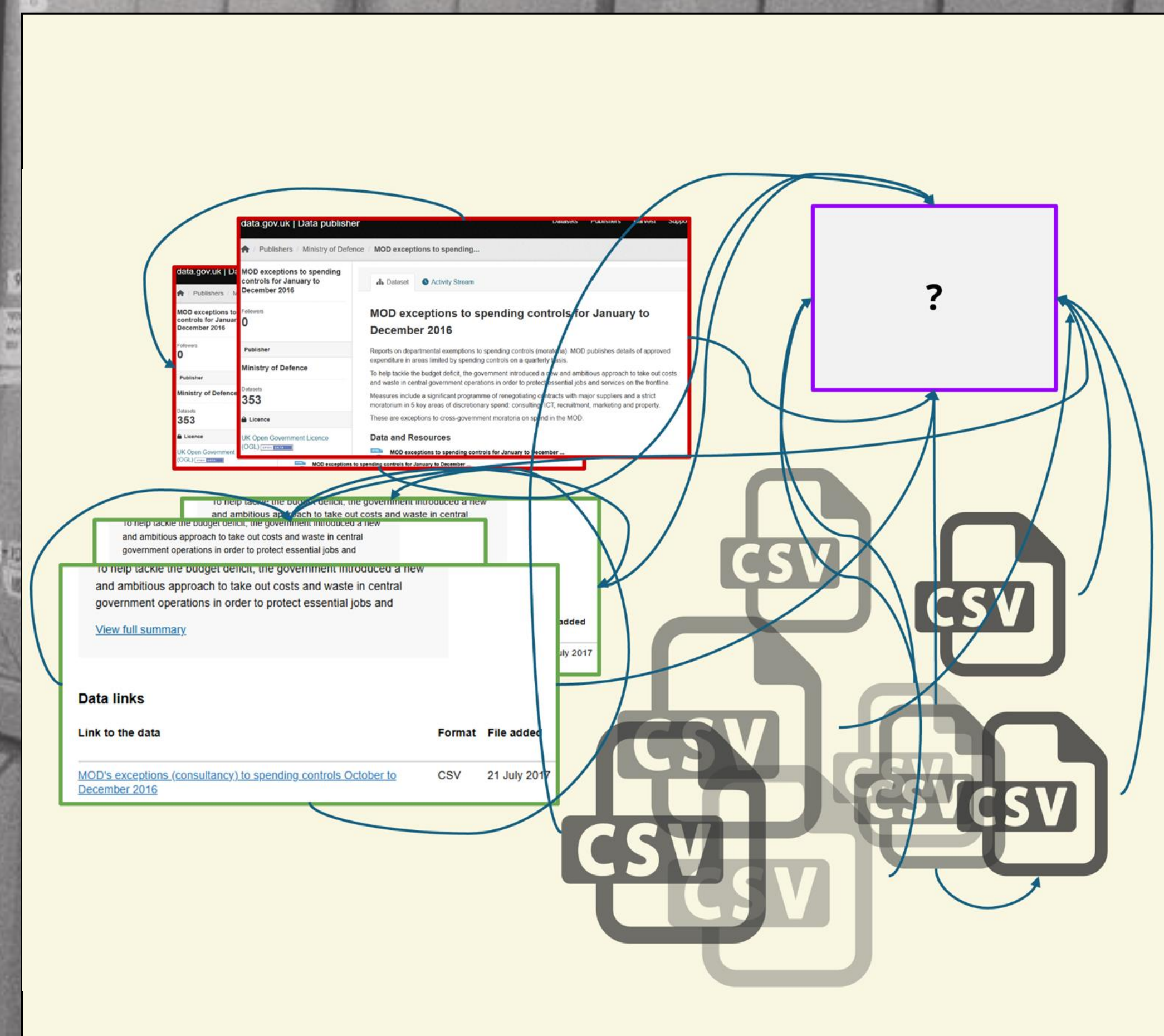


Let's *do*WARC!

Semantic Modelling of web archiving artefacts to support their sustainable long-term preservation



- Because of the distributed and dynamic nature of living web content, web archives struggle to implement sustainable and accessible approaches to versioning and context-anchoring of the archived data objects.
- When web archives are built through snapshotting of portions of the networked web, unique content is often interspersed with duplicate, or near-duplicate, material.
- This process can reduce the visibility of the relationship between digital objects within these collections, causing a loss of contextualisation that hinders meaningful description.
- For these reasons, web archiving practices can present challenges to digital preservation principles such as self-containment and self-description.

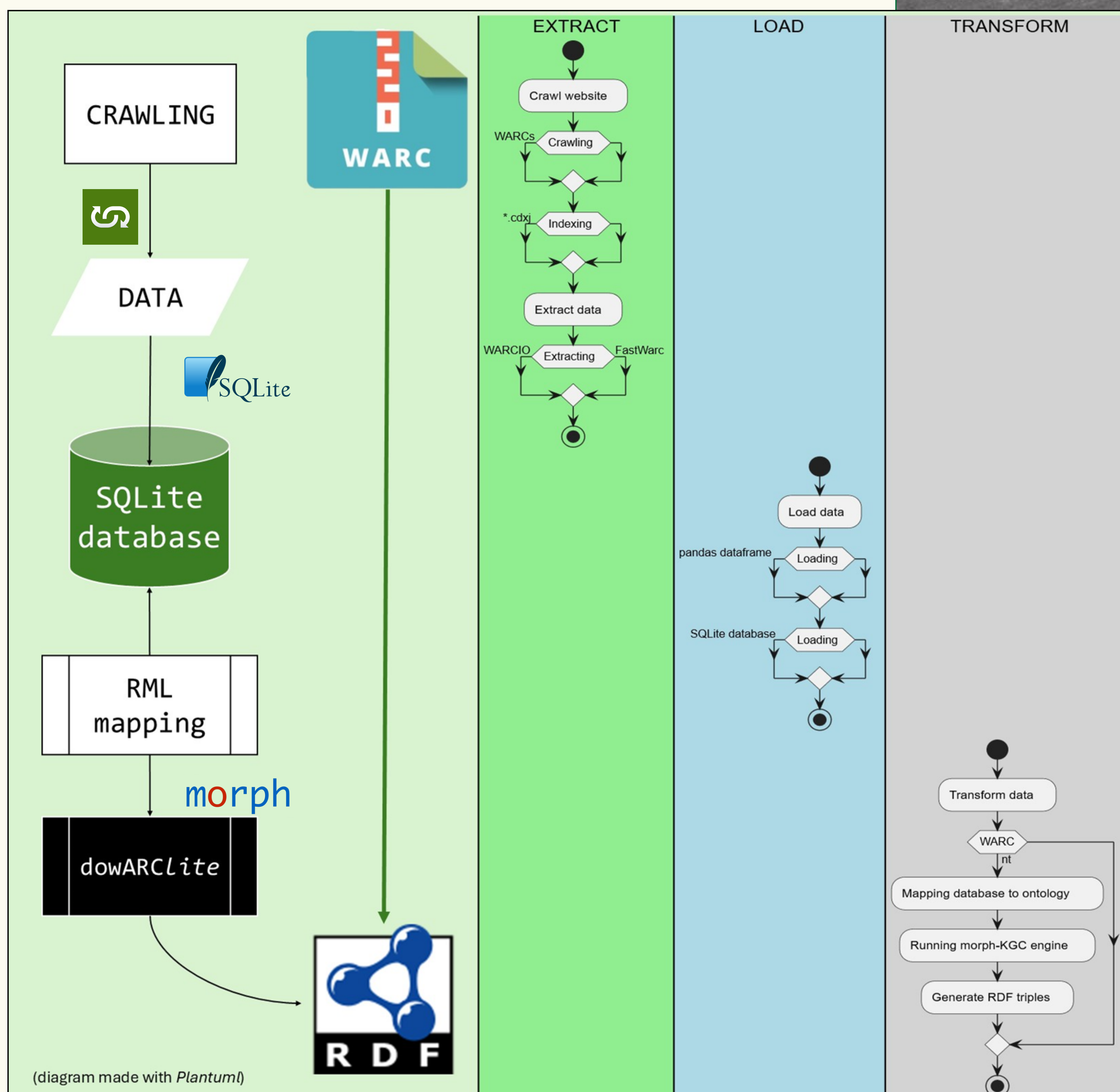


RDF modelling can effectively mitigate context-related ambiguities at scale, over ever-growing collections of WARC and derivative files



Modelling WARC files in RDF: the *do*WARC ontology

- Using the Resource Description Framework (RDF), we experimented with a local, small-scale implementation in a Jupyter environment
- We used standard web archiving tools (Browsertrix-crawler, WARCIO, FastWARC) to extract data from configuration files, WARC files, CDX files; we used our ISO-WARC-compliant *do*WARC ontology to describe it.
- We represented web archiving artefacts as interlinked RDF graphs, formalising relationships between resources, to express the context in which they were captured.
- We enriched the archive files with Semantic preservation metadata and linked them to the graph, which can be stored in the OAIS AIP as a machine-actionable representation.



This project is supported by a TNA-RLUK Professional Fellowship



THE
NATIONAL
ARCHIVES